

Probability and Statistics using R

Lab Session-2

Data Objects, Vectorization, Functions and Data Frames

Course Code : MAT445

Date:23.07.2026

Time : 3:15PM - 4:15 PM

Programming Tasks

1. Create numeric, character, and logical variables. Display their values and data types.
2. Create a list containing a student's Name, Age, CGPA, and Department. Access each element separately.
3. Create two vectors of 10 numbers each. Perform element-wise addition, subtraction, multiplication, and division.
4. Given a vector of marks, add 5 grace marks to every student using vectorized operations and display the updated marks.
5. Create a vector of numbers from 1 to 20. Display only the even numbers using vectorized logical indexing.
6. Write a user-defined function to calculate the area of a circle.
7. Write a function that accepts a vector and returns its maximum, minimum, and average values.
8. Write a recursive function to calculate the factorial of a given number.
9. Create a data frame containing Student ID, Name, Marks, and Department. Display students scoring more than 80 marks.
10. Create a data frame of employees with Name, Salary, and Experience. Add a new column named Bonus equal to 10% of the salary.